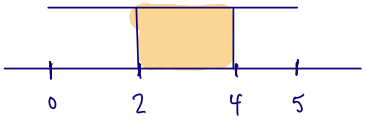
A

1. (b) probability = Area = $(2)\left(\frac{1}{5}\right) = 0.4$.



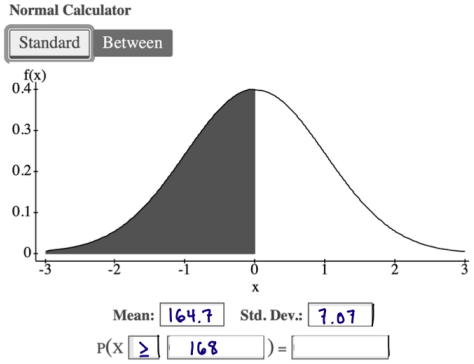
2. (a) appliances sold

   (b) time until scrapped

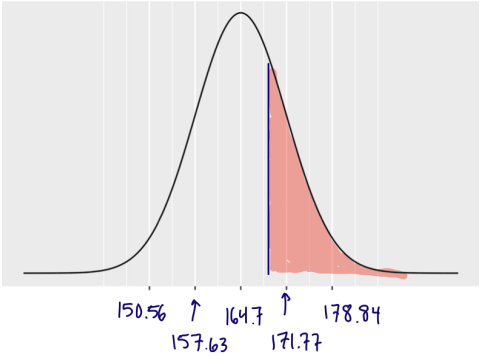
   (c) $H_0: \mu = 15$ vs. $H_a: \mu < 15$.

3. (a) III (b) II (c) IV (d) I

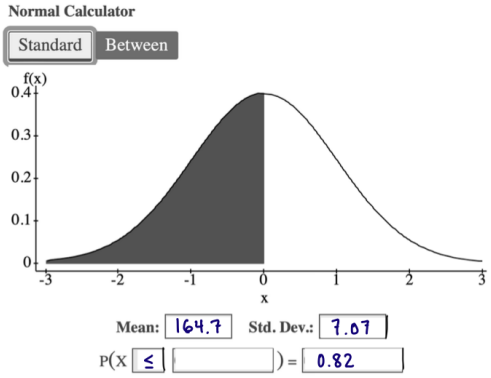
4. (a)



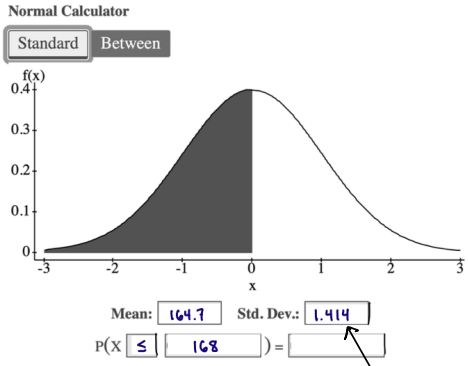
(b)



(c)



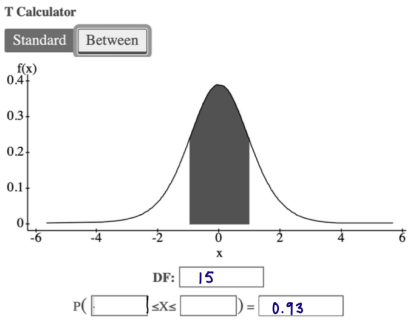
(d)



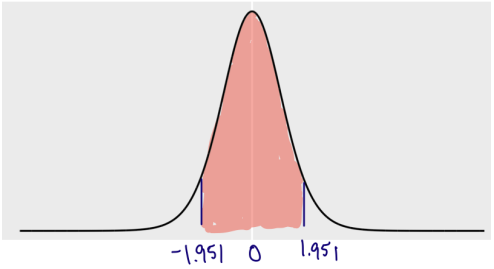
$\sigma = 7.07$ and so $SE_{\bar{x}} = \frac{7.07}{\sqrt{25}} = 1.414$

6. (a) μ is the average weight of these oranges considering the entire population.

(b)



(c)



(d) $149.2 \pm (1.95094)\frac{9.2}{\sqrt{16}} = 149.2 \pm (1.95094)(2.3) = 149.2 \pm 4.487$

or $(144.71, 153.69)$

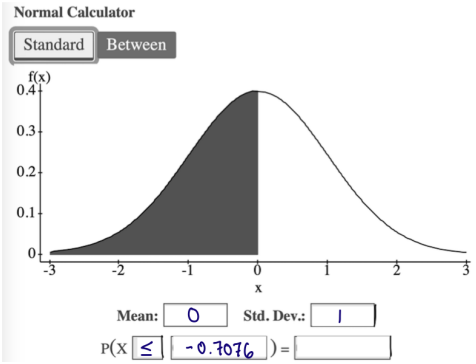
(e) Option (i)

7. (a) $\hat{p} = \frac{33}{72} \doteq 0.4583$

(b) It is valid, since $np = (72)(0.5) = 36$ and $n(1-p) = (72)(0.5) = 36$. Both are at least 5.

(c) $z = \frac{\hat{p} - p}{SE_{\hat{p}}} = \frac{0.4583 - 0.5}{\sqrt{(0.5)^2/72}} = \frac{-0.0417}{0.05893} \doteq -0.7076$

(d)



(e) Since the p-value is larger than $\alpha = 0.05$, we fail to reject the null hypothesis. Our data is consistent with the possibility that half of city residents support having no restrictions.